



Chapter Notes: Health: The Ultimate Treasure

Table of Contents

1. Health: Is It More Than Not Falling Sick?
2. How Can We Stay Healthy?
3. How Do We Know That We Are Unwell?
4. Diseases - What Are the Causes and Types?
5. How to Prevent and Control Diseases?

[View More](#) 

Have you ever wondered what it really means to be healthy-is it just not being sick, or is there more to it? This chapter will help you discover how good health includes not only your body, but also your mind, your habits, and your relationships with others. Let's explore how to stay healthy and keep diseases away in simple, practical ways!



An infographic featuring a woman in a yoga pose at the center. She is surrounded by various health and wellness icons: a heart with a pulse line, a purple lotus flower, a large grey 'X' mark, a red circle with a slash over a cigarette, a red circle with a slash over a burger and drink, a red circle with a slash over a bottle, a large orange alarm clock, a green apple, a green circle with '8h' indicating sleep, two pink dumbbells, a blue water bottle labeled 'WATER', a yellow banana, a purple scale showing '55', and a large orange carrot. The background is light blue with a faint circular pattern.

A healthy person can:

- Do tasks efficiently.
- Handle tough situations well.
- Get along with friends and others in society.

Key Aspects of Health

Physical Health: Taking care of your body through proper food, exercise, and sleep.

Mental Health: Staying positive and managing stress.

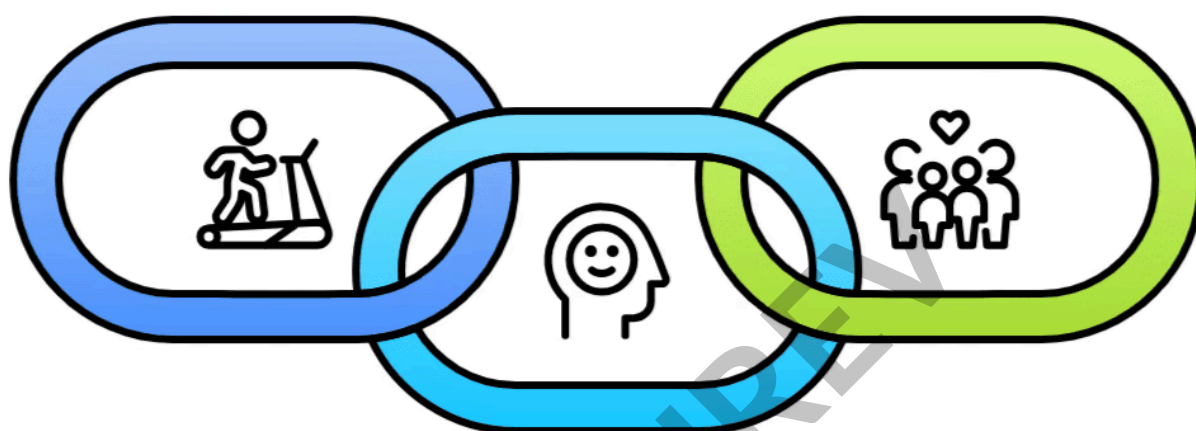
Social Health: Having good relationships with friends and others.

Physical Health

Focuses on bodily care through diet, exercise, and rest.

Social Health

Highlights the importance of relationships.



Mental Health

Emphasizes positivity and stress management.

A Student's Story

A Class 8 boy moved to a new city and school. He felt **lonely** with no friends and busy parents. He spent more time on his phone and social media, which made him feel worse and caused:

Headaches.
Weight loss.
Trouble sleeping.

Cause and Solution

A doctor suggested less screen time and seeing a counsellor. The counsellor helped him make friends, and his health improved.

Loneliness and too much screen time caused both physical and mental health problems.
Making new friends and support from adults helped him feel better.

Ayurveda and Health



Ayurveda teaches that health is a balance of body, mind, and surroundings.

Follow a **daily routine (dinacharya)** and **seasonal routine (ritucharya)**.

Eat fresh, wholesome food suited to your **prakriti** (body type).

Practice regular exercise, cleanliness, restful sleep, and a calm mind through **yoga**, **meditation**, and **mindfulness**.

How Can We Stay Healthy?

Staying healthy involves:

Eating **nutritious food**.

Keeping **clean** (personal hygiene and surroundings).

Exercising regularly.

Getting enough **sleep**.

Spending time with **family and friends**.

Having a **positive attitude**.

Good and Bad Habits

Good Habits:

Keep yourself clean and maintain personal hygiene.

Eat a healthy and balanced diet.



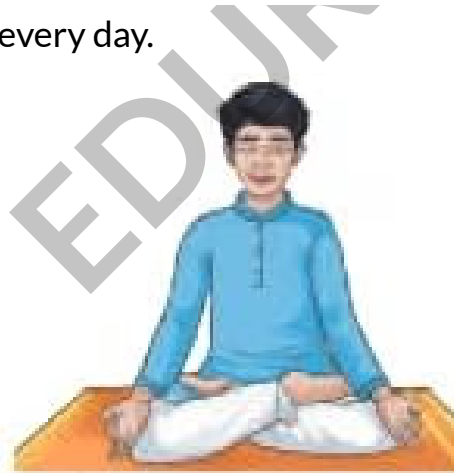
Eat a balanced diet

Exercise regularly.



Stay physically active

Make time to relax or meditate every day.



Manage stress

Bad Habits (harmful to health):

Spending too much time on mobile phones or screens.

Eating fast food or junk food every day.

Sleeping very late or not getting enough sleep.

Skipping meals, especially breakfast.

Maintain a Healthy Lifestyle

Our health depends on many factors. These factors include our lifestyle (how we live) and our environment (our surroundings).

We should Eat a **balanced diet** with: Plenty of fruits, vegetables, and whole grains.



Balanced Diet

Avoid processed, fatty, or sugary foods and drinks.

Stay **physically active** and Play outdoors, walk, run, cycle, or exercise.

Limit screen time and spend more time in nature.

Get enough **sleep** to help our body and mind rest.



Get enough sleep

Practice **yoga** or **pranayama** (simple breathing exercises) regularly.

Say '**NO**' to harmful substances like tobacco, alcohol, and addictive drugs.



*Say no to smoking or
alcohol*

Keep the Environment Clean

A clean, well-maintained playground is better for playing because it's safe and healthy. A dirty, polluted playground has flies, mosquitoes, and can make you sick.

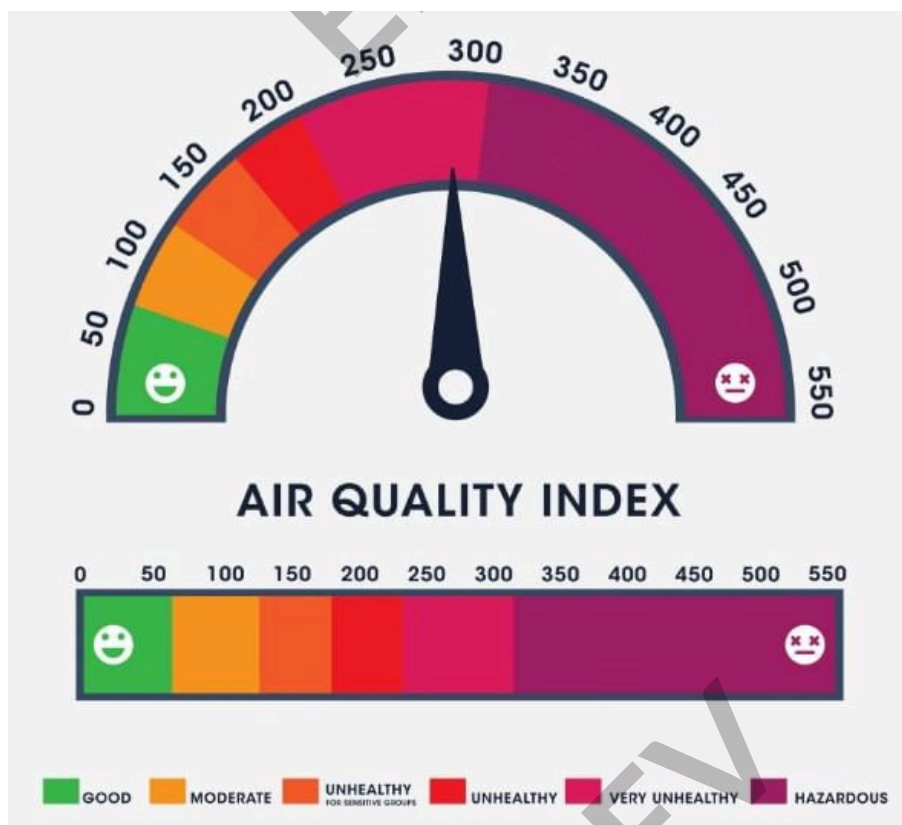
Why Clean Surroundings Matter:

Clean air and water are important for health.

Pollution from vehicles or factories can cause coughing or asthma.

The **Air Quality Index (AQI)** shows how clean or polluted the air is.

A clean environment helps you stay healthy and feel good.



Keeping Surroundings Clean:

Avoid littering and keep your area tidy.

A clean environment reduces the risk of diseases.

Importance of Feelings and Relationships

Health is not just about the body; **feelings** and **relationships** matter too.

Even with good food and a clean place, you may feel bad if you're lonely or upset.

Spend time with friends and family, talk, laugh, and have fun to keep your **mind healthy**.

MULTIPLE CHOICE QUESTION

Try yourself: What is one way to maintain a healthy lifestyle?

A Skip breakfast often

B Stay indoors all day

C Exercise regularly

D Eat fast food every day

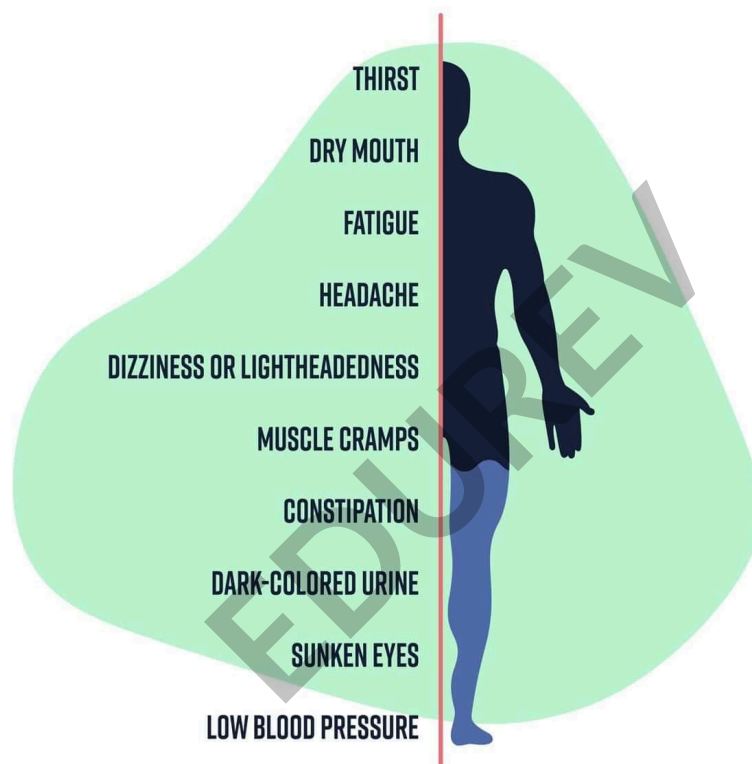
[View Solution](#)

How Do We Know That We Are Unwell?

Our body usually works in a specific way to keep us healthy.

When we feel **unwell**, something inside our body may not be working properly.

We experience **symptoms** and **signs** that show we are not healthy.



Symptoms of Dehydration

Symptoms and Signs

They act as clues to show that our body is not working as it should.

Doctors use these clues to understand the cause of our illness and suggest treatment.

Symptoms:

What we **feel** when we are unwell.

Examples: Pain, tiredness, dizziness.

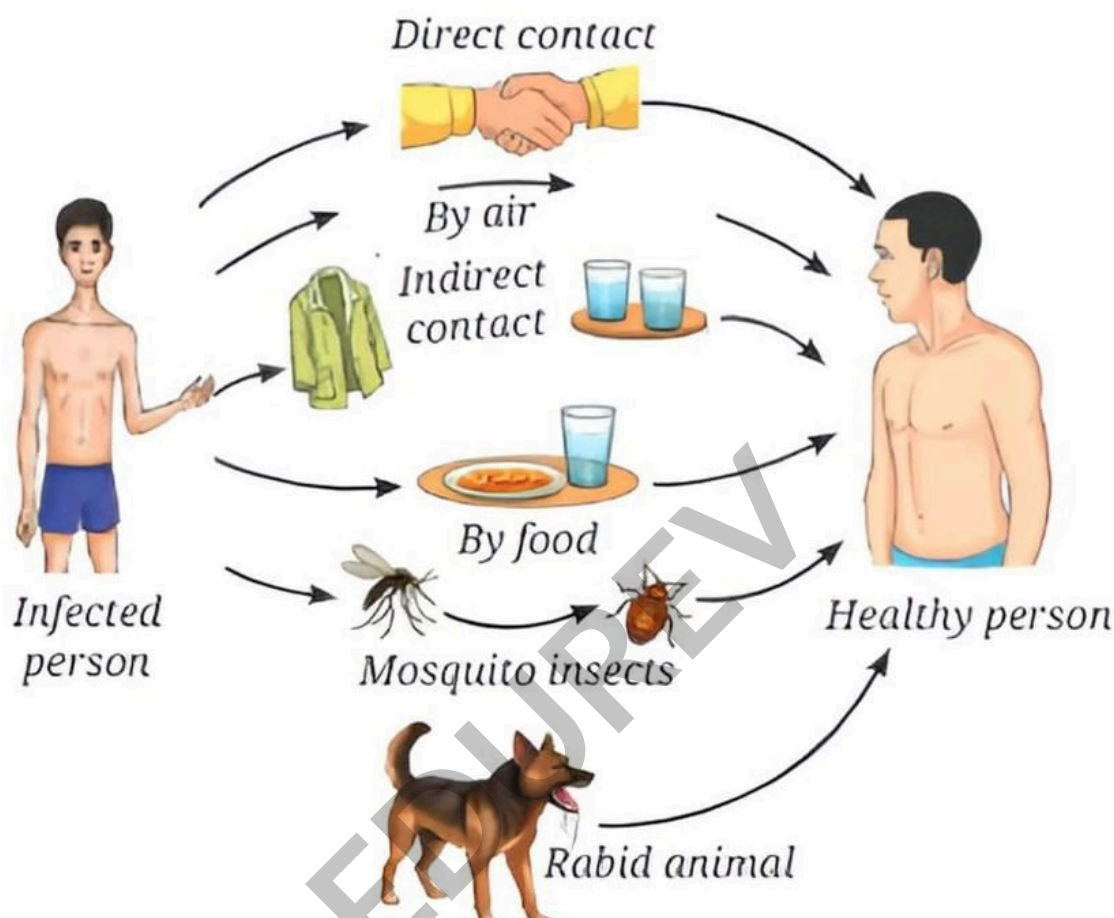
Signs:

Things that can be **seen** or **measured** by others, like a doctor.

Examples: Fever (high body temperature), rash, high blood pressure, swelling.

Diseases - What Are the Causes and Types?

A disease is a condition that stops the body or mind from working normally.
It happens when one or more organs or organ systems do not function properly.



Common Methods of Transmission of Diseases

Causes of Diseases

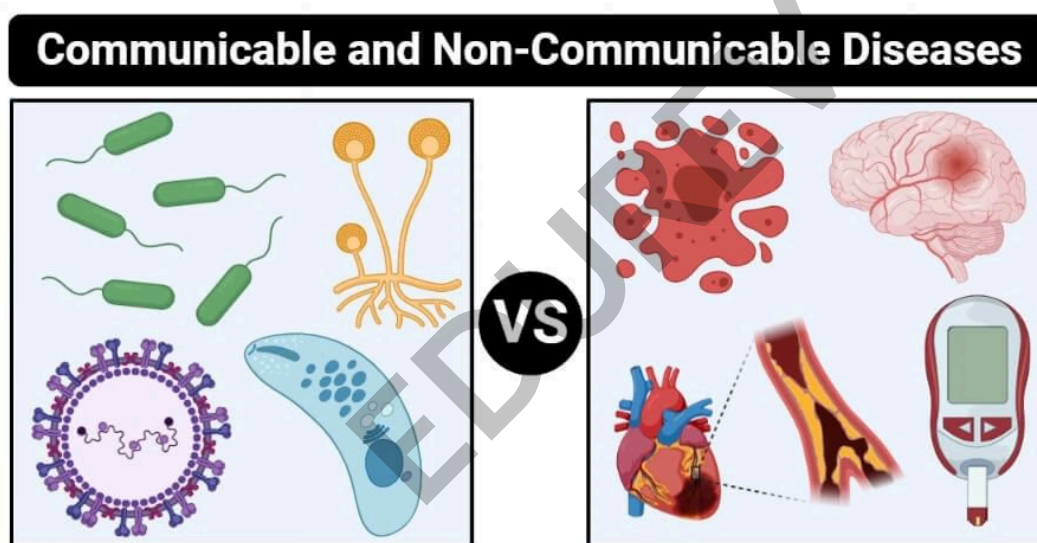
Pathogens: Germs like bacteria, viruses, fungi, worms, or protozoa (single-celled organisms) that cause diseases.

Poor Nutrition: Not eating enough healthy food can lead to diseases.

Unhealthy Lifestyle: Bad habits like lack of exercise or poor diet can cause diseases.

Types of Diseases

Diseases can be grouped into two major types based on their causes and how they spread:



1. Communicable Diseases: Can **spread** from one person to another. Caused by **pathogens**. **Examples:** Typhoid, dengue, flu, chickenpox, COVID-19.

Spread through:



Communicable Diseases

Air: Coughing or sneezing (e.g., flu, tuberculosis).

Direct contact: Touching an infected person (e.g., shaking hands).

Indirect contact: Sharing items like towels or handkerchiefs.

Contaminated food or water: Eating or drinking unsafe food/water (e.g., cholera, typhoid).

Vectors: Insects like mosquitoes or houseflies (e.g., malaria, dengue).

2. Non-Communicable Diseases (NCDs): Not caused by pathogens and **do not spread** from person to person. Linked to **lifestyle, diet, or environment**. **Examples:** Cancer, diabetes, asthma, heart disease.



Non- Communicable Diseases

Deficiency diseases: Caused by lack of nutrients (e.g., scurvy, anaemia, goitre).

Often **chronic** (last more than 3 months).

NCDs are Common in India due to:

Eating more processed food.

Less physical activity.

Longer lifespans.

How Are Communicable Diseases Caused and Spread?

All **communicable diseases** are caused by **pathogens**. These pathogens can enter our body through the air we breathe or by consuming contaminated food or water and more.

Diseases Spread Through Air: Diseases that are transmitted when pathogens in tiny droplets are released into the air by coughing, sneezing, or talking, and then breathed in by others.

Diseases	Causal agent	Site of infection	Symptoms	Preventive measures
Diseases spread through the air				
Common cold and influenza 	Virus	Respiratory tract	Nasal congestion and discharge, sore throat, fever, cough, body ache	Washing hands frequently, not sharing personal items, covering the mouth and nose
Chickenpox 	Virus	Respiratory tract, skin	Mild fever, itchy skin, rashes, blisters	Complete isolation of the patient, covering the mouth and nose, vaccination
Measles 	Virus	Skin, respiratory tract	Fever, sore throat, and reddish rashes on the neck, ears and other parts of the skin	Isolation of the patient, covering the mouth and nose, maintaining good hygiene, vaccination
Tuberculosis (TB) 	Bacteria	Lungs	Cough, fever, fatigue, loss of appetite, night sweats	Avoiding close contact with TB-infected people, covering the mouth and nose, maintaining good hygiene, getting vaccinated

Diseases Spread Through Contaminated Water/Food: Diseases that occur when people consume water or food contaminated with pathogens like bacteria, viruses, or parasites.

Diseases spread through contaminated water and food				
Hepatitis A 	Virus	Liver	Fatigue, fever, loss of appetite, nausea, vomiting, jaundice, pain in the upper right abdomen	Drinking boiled water, vaccination
Cholera 	Bacteria	Intestine	Diarrhoea and dehydration	Maintain personal hygiene and good sanitary habits, consumption of properly cooked food and boiled drinking water, vaccination
Typhoid 	Bacteria	Intestine	Headache, abdominal discomfort, fever, and diarrhoea	Maintain personal hygiene and good sanitary habits, consumption of properly cooked food and boiled drinking water, vaccination
Ascariasis (roundworms) 	Worms	Intestine	Worms in stool, loss of appetite, poor growth, diarrhoea, weight loss, anaemia	Maintain personal hygiene and good sanitary habits, consumption of properly cooked food and boiled drinking water

Diseases Spread by Insects (Vectors): Diseases that are passed on when insects such as mosquitoes or flies carry pathogens from one host to another, causing infection.

Diseases transmitted by insects				
Malaria 	Protozoa	Skin, blood	High fever, profuse sweating, periodic chills	Use of mosquito nets and repellents, wearing long-sleeved clothes, control of mosquito breeding in and around your home
Dengue fever (Break bone fever) 	Virus	Skin, blood	Fever, headache, muscle and joint pain, nausea	Use of mosquito nets and repellents, wearing long-sleeved clothes, control of mosquito breeding in and around your home, avoiding areas with still water

Prevention of Communicable Diseases

Keep yourself and surroundings **clean**.

Practice **good hygiene** (e.g., wash hands with soap).

Cover **mouth and nose** when coughing or sneezing.

Wear a **mask** in crowded places.

Avoid sharing personal items (e.g., towels, handkerchiefs).

Keep food and water **clean**.

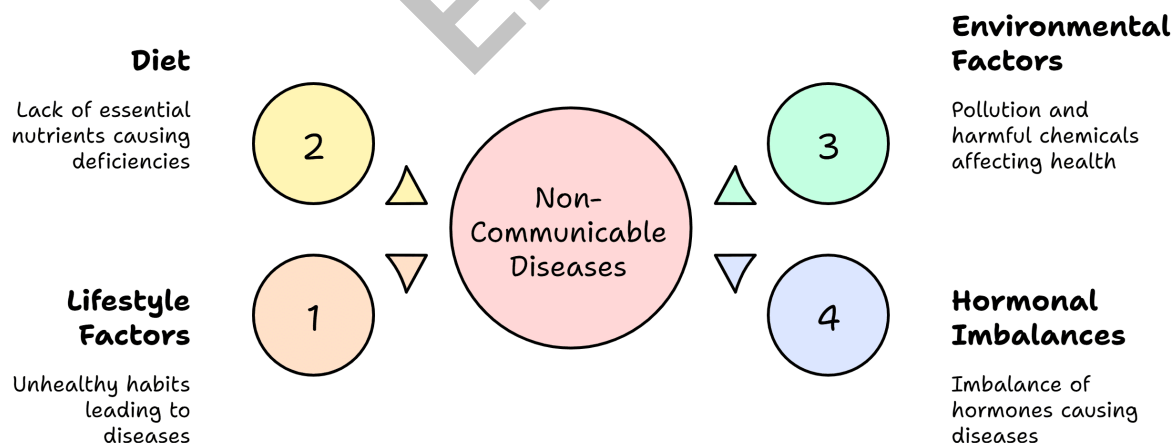
Stay home and rest when unwell to recover and avoid spreading disease.

Use **mosquito nets/repellents** and control insect breeding.

How Are Non-Communicable Diseases Caused?

Non-communicable diseases (NCDs) like cancer, diabetes, and asthma are **not caused by infections**. Instead, they are linked to factors such as lifestyle, diet, environment, and body functions:

Factors Contributing to Non-Communicable Diseases



Lifestyle Factors: Unhealthy habits like eating too much junk food, not exercising, smoking, or excessive stress can lead to diseases such as diabetes, heart disease, or some cancers.

Diet: Not getting enough essential nutrients can cause deficiency diseases (e.g., scurvy from lack of vitamin C, anaemia from lack of iron, goitre from lack of iodine). These are also non-communicable.

Environmental Factors: Pollution, exposure to harmful chemicals, or living in unhealthy surroundings can contribute to diseases like asthma or cancer.

Hormonal Imbalances: Sometimes, diseases like diabetes happen due to imbalance of hormones in the body.

Chronic Nature: NCDs usually last a long time (more than 3 months) and need ongoing care.

S. No.	Name of Common Lifestyle-Related Diseases	Signs and Symptoms	Suggested Lifestyle Change(s)
1.	Obesity	- Excess body weight (BMI ≥ 25 kg/m² for Asians, as per WHO)	- Eat a balanced diet with fruits, vegetables, and whole grains. - Exercise regularly (e.g., 30 minutes, 5 times a week). - Limit high-calorie, processed, and sugary foods.
2.	Diabetes	- Frequent urination - Excessive thirst - Weight loss - Tiredness - Slow healing	- Follow a diabetes-friendly diet with low-sugar, complex carbohydrates. - Engage in regular physical activity (e.g., walking, jogging, yoga). - Monitor blood sugar levels regularly. - Take prescribed medications or insulin as advised by a doctor.
3.	High Blood Pressure (Hypertension)	- Often no symptoms (silent killer) - Possible headaches, dizziness, or nosebleeds in severe cases	- Reduce salt intake (e.g., follow DASH diet). - Exercise regularly to maintain <u>healthy</u> weight. - Avoid smoking and limit alcohol consumption. - Manage stress through relaxation techniques like yoga or meditation.
4.	Heart Disease	- Chest pain - Shortness of breath - Fatigue - Irregular heartbeat	- Eat a heart-healthy diet with fruits, vegetables, whole grains, and lean proteins. - Avoid smoking and excessive alcohol. - Exercise regularly (e.g., 30 minutes most days). - Monitor cholesterol and blood pressure regularly.

How to Prevent and Control Diseases?

You may have heard the saying: "**Prevention is better than cure.**" This means it's easier and wiser to stop a disease from happening than to try to cure it after you get sick.

Good sanitation and cleanliness are some of the simplest and most effective ways to reduce the spread of diseases, especially those that spread from person to person or through contaminated water and food. **Example:**

Odisha - community-led sanitation campaign



In Bhadrak district, Odisha, a community campaign encouraged people to build and use toilets.

As more families stopped open defecation, child health improved, and diseases like diarrhoea and infections greatly reduced.

Simple steps like building and using toilets, keeping surroundings clean, and good waste disposal **prevent germs from spreading**.

Community action can have a big impact on public health-when people work together, the whole area becomes healthier.

Ability of the body to fight diseases

Why do some people get sick more often than others, even in the same environment?

The ability of the body to fight diseases is called **immunity**.

Our **immune system** is a group of special cells, tissues, and organs that recognise and fight off harmful germs (pathogens) that can make us sick.

Role of Vaccines

Vaccines are special injections or drops that help your immune system become stronger against specific diseases.



They do this by "teaching" your body how to recognise and destroy certain germs before they can make you sick. This is called **acquired immunity**.

Some vaccines use tiny, harmless parts of the germ (or a dead/weakened germ) to "train" your immune system.

If you're exposed to the real germ in future, your body is ready and can fight off the disease quickly.

Tetanus shot: Contains a safe (inactive) form of a bacterial toxin, which helps your immune system learn how to fight tetanus without making you actually sick.

Other common vaccines: Polio, measles, hepatitis, and more-many are given in childhood.

Edward Jenner and the smallpox vaccine

Edward Jenner and smallpox vaccine: In the late 1700s, Jenner realized that people who got cowpox didn't get smallpox. He created the first vaccine by using material from cowpox sores. This discovery eventually helped *eradicate* smallpox worldwide.

In olden days, India also had a method called **variolation**, where a bit of material from a mild smallpox infection was used to protect healthy people.

Why are vaccines important?

Vaccines prevent diseases: They are used before you get sick, to stop serious illnesses from starting at all.

Community protection: When more people get vaccinated, diseases have less chance to spread-so even those who can't get vaccinated are protected.

Vaccines are safe and tested: Scientists and doctors check them carefully before use.

India's Role in Vaccines

India is a major global vaccine producer, supplying vaccines to many countries.

Indian vaccine companies played a key role during the COVID-19 pandemic and continue to support global health efforts

MULTIPLE CHOICE QUESTION

Try yourself: What is the main focus of the text?

- A Improving technology
- B Studying animal behavior
- C Developing new medicines
- D Preventing and controlling diseases

View Solution

Treatment of Diseases

If our **immune system** cannot fight off infection, we fall ill and need help from a doctor. The doctor gives us **medicines**, and sometimes these are called **antibiotics**.

What are antibiotics?

Antibiotics are special medicines that **kill bacteria** in our body.

They help cure diseases caused by bacteria, such as tuberculosis and typhoid.

Important: Antibiotics **do NOT work** against diseases caused by **viruses** (like colds, flu, or COVID-19) or **protozoa**.

Discovery of Antibiotics

In **1928**, Alexander Fleming, a scientist in London, discovered the first antibiotic called **penicillin**.

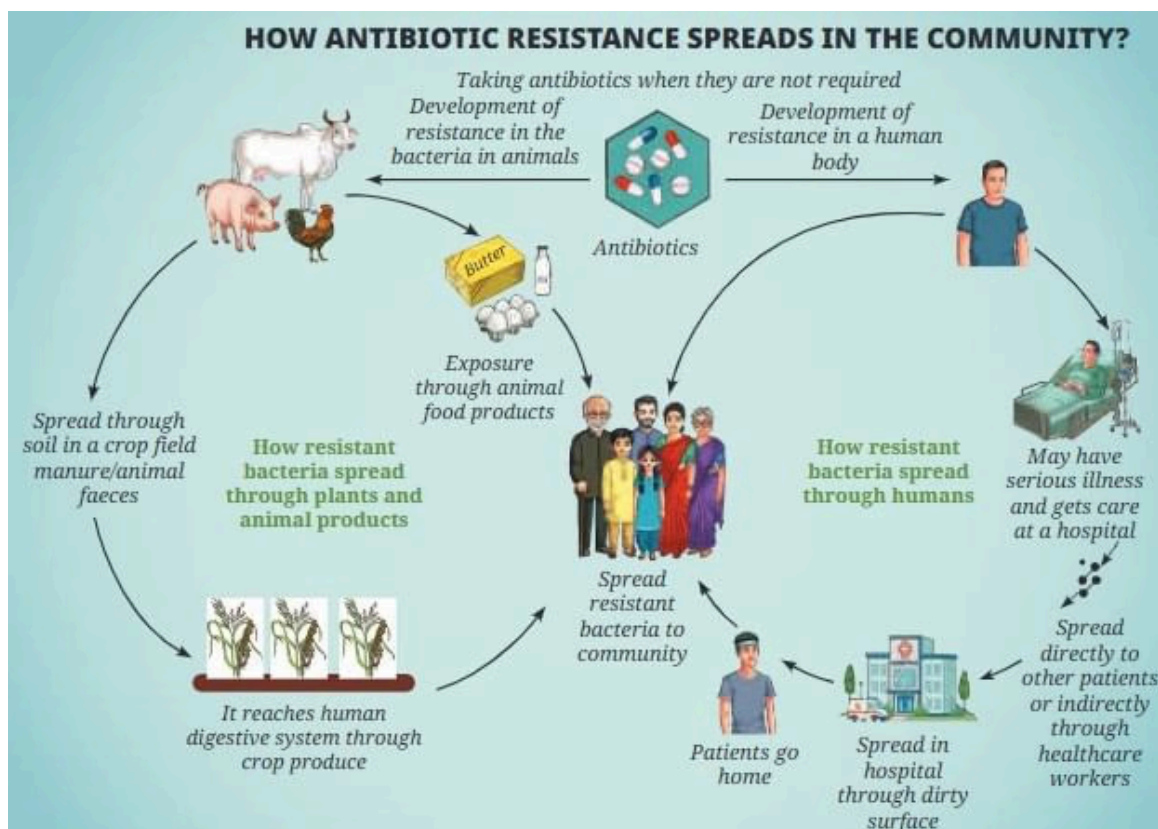
He found that a mould on a petri dish killed harmful bacteria growing there. This mould made a substance (penicillin) that stopped bacteria from growing.

Since then, antibiotics have saved millions of lives.

What is Antibiotic Resistance?

Antibiotic resistance happens when bacteria change and **stop being killed by antibiotics** that used to work.

This makes infections **much harder to treat** and can cause longer illnesses and more complications.



How does antibiotic resistance develop and spread?

Taking antibiotics when we **don't need them** (like for viral infections).

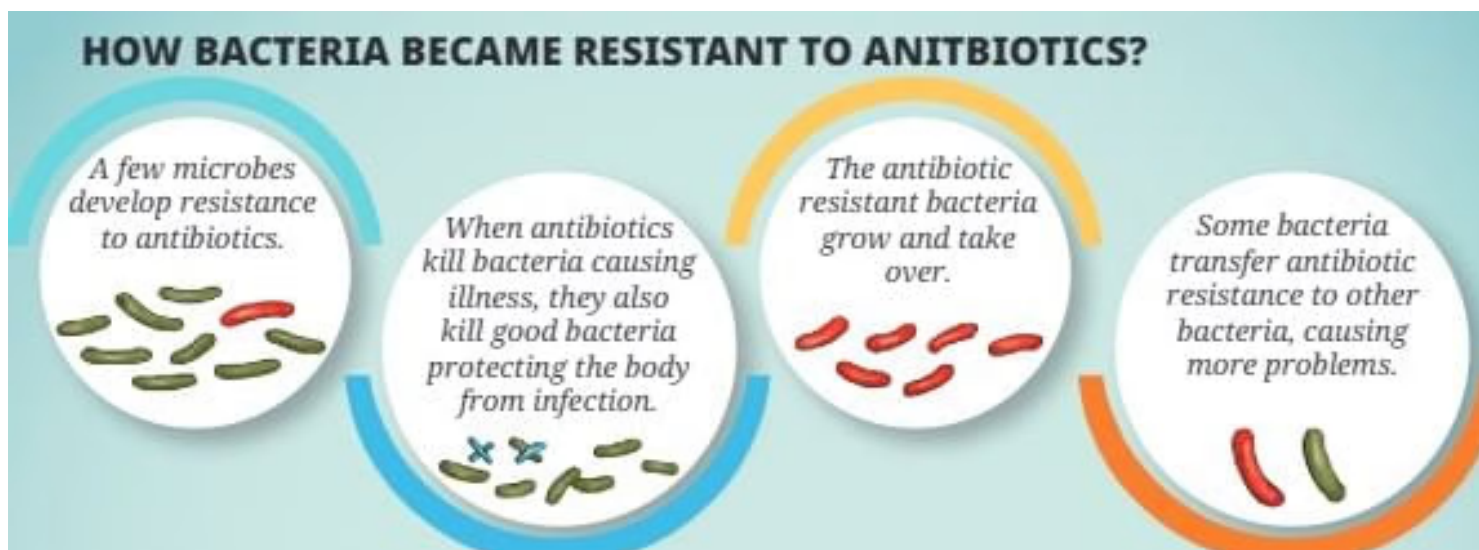
Not finishing all the doses as prescribed by the doctor.

Overuse of antibiotics in animals and plants.

Spread can happen through people, animals, food, water, and soil.

If you take antibiotics for a sore throat caused by a virus, bacteria in your body may become resistant to antibiotics.

When farmers give antibiotics unnecessarily to animals (like cows), resistant bacteria can spread through meat, milk, or even through the soil.



How to Prevent Antibiotic Resistance?

Take antibiotics **only when prescribed by a doctor**.

Always finish the **full course** of medicine, even if you feel better.

Never use leftover antibiotics or someone else's prescription.

Avoid buying antibiotics without a doctor's prescription.

Farmers should avoid giving antibiotics unnecessarily to animals.

Remember: Using antibiotics wisely keeps them working for everyone!

Traditional Medicine Systems

Systems like **Ayurveda, Siddha, and Unani** use natural remedies (herbs, oils, minerals) and focus on good food, exercise, and overall healthy living.

They can be helpful in treating simple health problems and for everyday wellness.

However, for some serious or advanced diseases, modern medicine may work better.

Managing Non-Communicable Diseases

Diseases like diabetes, cancer, and heart disease are managed by:

Taking medicines (as advised by doctors)

Making healthy lifestyle changes (diet, exercise)

Rehabilitation (special help to get better)

Early diagnosis and regular check-ups are very important.

Key Points to Remember

Health means complete physical, mental, and social well-being-not just being free from diseases.

Happiness and health go together: Being happy can help you stay active and healthy, and good health can improve your mood.

A disease is any condition that stops the body or mind from working normally.

Symptoms are feelings we notice ourselves (like pain or tiredness); **signs** are visible or measurable changes (like fever, rash, or swelling).

Non-communicable diseases (like diabetes, cancer, and heart disease) are not caused by germs, but by lifestyle factors and environment. They can be prevented by eating healthy, staying active, and avoiding harmful habits.

Communicable (Infectious) diseases are caused by **pathogens**-germs such as bacteria, viruses, fungi, or worms-and can spread from person to person.

Good hygiene and clean surroundings are essential to prevent many diseases.

Our immune system is the body's natural defense that fights harmful germs and keeps us safe.

Vaccines help our immune system "learn" how to fight certain diseases and provide protection before we actually get sick.

Early diagnosis and proper treatment help manage, control, or cure many diseases.

Healthy habits-like regular exercise, balanced diet, enough sleep, and positive relationships-keep both our body and mind strong.

Antibiotics treat bacterial infections only, not viral ones. Overuse can cause antibiotic resistance-so take them only as prescribed.

Prevention is better than cure: Practising good habits, getting vaccinated, and staying informed are the best ways to maintain health.